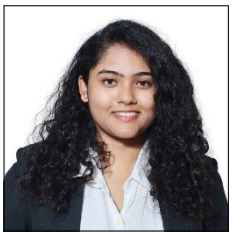
Vaidarbhi Kailas Buddhe

Lilai, Malati Nagar, Near Zudio, Khat Road, Bhandara, Maharashtra – 441904

+91 8484968019 | imt22vs.buddhe@stumarj.ictmumbai.edu.in

linkedin.com/in/vaidarbhi-buddhe | https://vaidarbhi-buddhe.github.io/Portfolio/



PROFESSIONAL SUMMARY

Final-year Integrated M.Tech Chemical Engineering student (CGPA: 8.73) with hands-on experience across refinery research, lubricant manufacturing, energy-efficiency projects, and process operations. Skilled in Aspen Plus, Aspen HYSYS, Python, and data analysis, with experience in catalyst development, process simulation, production supervision, and continuous improvement. Interested in Process Engineering, Process Optimisation, Manufacturing, and Energy & Petrochemical Operations.

EDUCATION

Institute of Chemical Technology (ICT Mumbai) Mumbai, Maharashtra
Integrated M.Tech., Chemical Engineering (Major); Minor: Petroleum and Petrochemical Engineering Expected Graduation: 2027
*CGPA: **8.73**/10.00 (Trimester 11)*

J. M. Patel College of Arts, Science and Commerce Bhandara, Maharashtra
H.S.C. (Science), Maharashtra State Board – 88.17% *2022*

Nutan Kanya High School Bhandara, Maharashtra
S.S.C., Maharashtra State Board – 100% *2020*

TECHNICAL KNOWLEDGE

Core Domain Knowledge: Process Optimization, Material & Energy Balances, Heat Transfer, Mass Transfer, Reaction Engineering, Catalyst Manufacturing & Batch Traceability, Process Safety

Laboratory & Analytical Techniques: HPLC, GC, pH Meter, Conductometer, UV Spectrophotometer, Viscosity Testing, Flash Point, Pour Point

Process Simulation Software: Aspen Plus, Aspen HYSYS, DWSIM

Programming & Data Analysis: Python, MATLAB, Minitab, MS Excel

Quality Systems & Energy Efficiency: Energy Efficiency Assessment, Heat-Loss Analysis, Sustainability Analysis, Continuous Improvement, Lean Six Sigma

Industrial Tools: SAP ERP Systems, PLC, CATIA V5

PROJECTS

Master's Thesis: Development of a Dynamic Model for Process Monitoring and Fault Detection

Major Guide: Dr Debashish Kundu [Mar 2026 - Present]
Minor Guide: Dr Supriyo Kumar Mondal

This thesis focuses on developing a process simulation framework integrated with AI/ML techniques to analyse dynamic plant behavior, identify operational anomalies, and support fault detection and diagnosis. The project aims to improve process reliability, operational efficiency, and energy performance through simulation-driven analysis and data-driven insights.

REASERCH/ INDUSTRIAL EXPERIENCE

Dow India Technology Centre – Project Intern Jul 2026 – Oct 2026
Navi Mumbai, Maharashtra
Guide: Dr. Neeraj Rohilla
***Project:** Data Interpretation for EOR Field Datasets: Decline Curve Fitting and Incremental Oil Determination*

- Analyzed production histories of **5 synthetic oil wells** at Dow India Technology Centre (DITC), selecting DITC-5 for decline curve analysis using engineering assessment and production trend interpretation.
- Developed and validated **exponential, harmonic, and hyperbolic decline models** using Python; selected the harmonic model based on the lowest blind-forecast errors (RMSE: 315.10, MAE: 292.06) to estimate **~527,000 barrels of apparent incremental oil**.

CSIR Indian Institute of Petroleum – Project Intern Nov 2025 – Feb 2026
Dehradun, Uttarakhand
Guide: Dr. Belal Haider
***Project:** Thermodynamic and Experimental Analysis for Syngas to LPG Conversion through Modified Route*

- Prepared Cu–ZnO–ZrO_2–Al_2O_3 (CZZA) catalysts via co-precipitation and evaluated product samples using Gas Chromatography (GC) for syngas-to-LPG conversion.
- Simulated a **27-reaction network in Aspen Plus (RGibbs)** and conducted sensitivity analysis across temperature, pressure, and CO/H_2 ratio to assess LPG formation.

Ultra Plus Lubes Pvt. Ltd – Chemical Engineering Intern April 2025 – Jun 2025
Panvel, Maharashtra

- Supervised a **12-member production team** across lubricant blending, filling, packing, and dispatch, with exposure to **PLC-controlled** and manual blending systems.
- Audited **daily production data** and tracked oil wastage across blending and filling stages to identify operational losses and support production-efficiency improvements.

Indorama Synthetics (IND) Ltd. – Internship Trainee Aug 2024 – Oct 2024
Butibori (Nagpur), Maharashtra
Project: *Enhancing Energy Efficiency in CF-HTM Systems Through Insulation Upgrades*

- Evaluated a proposed insulation upgrade from mineral wool to calcium silicate for CF-HTM systems, assessing surface temperature, heat loss, energy consumption, and emissions.
- Assessed projected reductions in surface temperature from **6°C to 3°C** and heat loss by **50%**; participated in plant-floor Kaizen initiatives and completed Lean Six Sigma Green Belt training.

Sanjay Techno Products – Chemical Engineering Intern Feb 2024 – Mar 2024
Chh. Sambhajinagar, Maharashtra

- Studied injection moulding of engineering and commodity polymers, examining process operations and common product defects.
- Analysed defects including flash, sink marks, burn marks, voids, jetting, and warpage to build understanding of polymer-processing quality issues.

Clarion Organics Ltd. – Internship Trainee Feb 2024 – Mar 2024
Tumsar, Maharashtra

- Studied pharmaceutical-intermediate manufacturing processes, including nitration, reduction, hydrogenation, filtration, centrifugation, and drying.
- Performed **Karl Fischer titration and factor determination** for moisture analysis and examined industrial equipment, including glass-lined/SS316 reactors, hydrogenators, centrifuges, sparkler filters, and dryers.

CERTIFICATION & AWARDS

Professional Certifications

- Aspen Plus® Simulation Software – A Basic Course for Beginners (NPTEL, IIT Guwahati) Elite + Gold Certification — Oct 2026
- Six Sigma White Belt (Council of Six Sigma Certifications) — Jul 2026
- Aspen Plus - Getting Started (ChemEngGuy) — Aug 2025
- Introduction to Symbolic Math with MATLAB (MATLAB) — Aug 2025

Awards & Certifications

- Vedvalli Vaidyanathan Girl Student Award for Academic Excellence (Institute of Chemical Technology Mumbai) — 2022 – 2023
- State Level Rugby Competition 2nd Place, Government of Maharashtra State — 2019 – 2020

POSITION OF RESPONSIBILITY

Off-Campus Representative – The Bombay Technologist, ICT MARJ 2025 – 2026
• Represent the institute externally by coordinating outreach initiatives and fostering engagement with students.

Deputy Head – Literature Club, ICT MARJ 2024 – 2025
• Led the planning and execution of literary events, competitions, and club activities while mentoring members and promoting creative expression.

Member – Margjal Editorial Board, ICT MARJ 2022 – 2024
• Contributed to content creation, editing, and publication of the institute's editorial magazine, ensuring high-quality and engaging material.

Event and Competition Team Member – Aakriti (Tech Fest), ICT MARJ 2023 – 2024
• Assisted in organising technical events and competitions by coordinating logistics, participant engagement, and smooth event execution.

Science Outreach Volunteer, Rainbow EduFest, Rotary Club of Jalna Feb 2023
• Contributed to a Rotary Club science outreach initiative that engaged 500+ students across 15 schools through interactive science demonstrations and awareness activities

ADDITIONAL INFORMATION

Languages: English, Hindi, Marathi

Interests: Journaling, Sketching, Travelling, Learning New Languages